

The Curvature-Responsive Vacuum: Theoretical Foundations and a Cosmological Realization

A Conceptual Framework and Cosmological Realization

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Abstract

This white paper presents a unified conceptual framework in which the vacuum is treated as a curvature-responsive medium with memory, and shows how this idea manifests both as a theoretical architecture and as a concrete, numerically validated cosmological model. The document is expository in intent: its purpose is to convey the underlying mechanism, the mathematical structure, and the phenomenological consequences in a single coherent narrative, and to be honest about what has been established and what remains open.

Part I develops the theoretical foundations of the *curvature-responsive vacuum*: the memory law, the fixed-point structure, the exact decoupling condition $Q(H_*) = 0$, the heteroclinic flow connecting an inflationary saddle to a late-time dark-energy attractor, and an illustrative auxiliary-field realization via a nonlocal curvature action. Part II presents the *Curvature-Responsive Vacuum Cosmology* (CRVC) model — a two-ODE phenomenological realization of this mechanism — together with its full numerical results: a non-monotonic S-curve growth history, a falsifiable peel-away epoch at $z^* \approx 0.65$, a suppressed clustering amplitude $S_8 = 0.750 \pm 0.020$ consistent with weak-lensing surveys, and a continuous degeneracy manifold in parameter space — the *Goldilocks Line*. Part III makes the conceptual and mathematical bridge between the two layers explicit, identifies what the full nonlocal theory must reproduce to recover CRVC phenomenology exactly, and maps the limits of the low-energy approximation.

The document closes with an honest accounting of open questions and a three-test viability program. It is a conceptual synthesis and a research program, not a finished theory.

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Executive Summary

Ontology. The vacuum is modeled as a curvature-responsive medium with memory: its energy density responds to the background spacetime geometry with a finite relaxation time, rather than remaining rigidly constant.

Mechanism. The vacuum energy density obeys a memory law driven by the Ricci scalar:

$$\dot{\rho}_{\text{vac}} + \mu(H) \rho_{\text{vac}} = \kappa R.$$

This produces exact decoupling at de Sitter fixed points, a heteroclinic flow from inflation to late-time acceleration, and — in a concrete low-energy realization — a non-monotonic growth history with a measurable imprint on the matter power spectrum.

Realization. The CRVC model implements this mechanism via two coupled ODEs and yields:

- a Λ -like background expansion ($w_{\text{vac},0} = -0.996$),
- a non-monotonic S-curve in $f\sigma_8(z)$,
- a peel-away epoch $z^* \approx 0.65$ (falsifiable),
- a suppressed clustering amplitude $S_8 = 0.750 \pm 0.020$,
- and a continuous viability manifold — the Goldilocks Line.

Status. The CRVC phenomenology is numerically validated. The theoretical foundations are established at the concept level. Ghost conditions, nonlinear structure formation, and a full Boltzmann-code implementation remain open.

How to Read This Document

This white paper is organized into three parts.

Part I develops the mechanism and mathematical architecture of the curvature-responsive vacuum. It is the theoretical layer: it does not require numerical results to stand on its own.

Part II presents the CRVC model as a concrete phenomenological realization. It contains the full numerical results and is largely self-contained for readers interested in the observational predictions.

Part III makes the bridge between the two layers explicit. Readers who want to understand why CRVC is not merely ad hoc, and what a deeper derivation would need to reproduce, should read this part.

Readers interested primarily in the mechanism should focus on Part I. Readers interested primarily in the phenomenology should focus on Part II. Readers interested in the conceptual architecture connecting the two should read Part III.

Part I

The Curvature-Responsive Vacuum: Theoretical Foundations

1 Motivation and Ontological Reframing

The standard cosmological constant treats the vacuum as a rigid, structureless background contribution to the stress-energy tensor. In this picture, the vacuum does not respond to curvature, does not evolve dynamically, and does not participate in the causal structure of the universe except through its constant energy density.

Several independent lines of reasoning suggest this picture is incomplete. Anomaly-induced effective actions generically contain nonlocal curvature terms whose stress-energy is time-dependent. Semiclassical backreaction in quantum field theory in curved spacetime produces memory kernels: the vacuum stress-energy at a given moment depends on the integrated history of the geometry. Horizon thermodynamics implies that vacuum energy is sensitive to the causal structure of spacetime, and renormalization-group arguments suggest a scale-dependent vacuum response to curvature. None of these is a modification of gravity; each is a consequence of taking seriously the quantum nature of the vacuum in a dynamical spacetime. These motivate a different organizing principle:

The vacuum is a curvature-responsive medium with memory. Its energy density responds to spacetime curvature with a finite relaxation time, rather than remaining rigidly constant.

This reframing is distinct from modified gravity, dark fluids, and running cosmological constants. It does not alter the Einstein gravitational sector. It does not introduce new propagating fields. It posits only that the vacuum energy density is dynamical, responding to the background geometry on a timescale set by the physics of the vacuum itself.

The remainder of Part I develops the mathematical structure that follows from this ontology.

2 The Memory Law

The central dynamical assumption is that the vacuum energy density obeys a relaxation-type equation driven by the scalar curvature:

$$\dot{\rho}_{\text{vac}} + \mu(H) \rho_{\text{vac}} = \kappa R. \tag{1}$$

Here κ is a curvature-to-vacuum coupling with dimensions of [energy density]/[R], $\mu(H)$ is a Hubble-dependent relaxation rate, and $R = 6(2H^2 + \dot{H})$ is the Ricci scalar in a spatially flat FRW universe.

The physical interpretation has three layers. First, the term κR is a curvature source: spacetime curvature continuously injects energy into the vacuum sector. Second, the term $\mu(H)\rho_{\text{vac}}$ is a relaxation drain: the vacuum energy relaxes back toward zero (or toward the curvature-driven equilibrium) at a rate set by $\mu(H)$. Third, the Hubble dependence of μ encodes the idea that the relaxation timescale is not universal but depends on the expansion rate — on the characteristic timescale of the background geometry itself.

It is useful to define the exchange term

$$Q(H) \equiv \mu(H) \rho_{\text{vac}} - \kappa R, \quad (2)$$

so that $\dot{\rho}_{\text{vac}} = -Q(H)$. When $Q > 0$, energy flows from the vacuum into the geometry; when $Q < 0$, energy flows from geometry into the vacuum. The sign of Q therefore determines whether the vacuum is charging or discharging at any given epoch.

An illustrative auxiliary-field realization of Eq. (1) from a nonlocal curvature action is presented in Section 6.

3 Fixed Points and Exact Decoupling

A de Sitter fixed point satisfies $\dot{H} = 0$ and $\dot{\rho}_{\text{vac}} = 0$. In a flat FRW universe, these conditions together with the Friedmann equation give

$$R_* = 12H_*^2, \quad \rho_{\text{vac},*} = \frac{3H_*^2}{8\pi G}.$$

Substituting into Eq. (1) and requiring $\dot{\rho}_{\text{vac}} = 0$:

$$\mu(H_*) = 32\pi G \kappa. \quad (3)$$

Proposition. *At any de Sitter fixed point satisfying Eq. (3), the exchange term vanishes exactly: $Q(H_*) = 0$.*

Proof. Substituting $R_* = 12H_*^2$ and $\rho_{\text{vac},*} = 3H_*^2/(8\pi G)$ into Eq. (2):

$$Q(H_*) = \mu(H_*) \cdot \frac{3H_*^2}{8\pi G} - \kappa \cdot 12H_*^2 = H_*^2 \left[\frac{3\mu(H_*)}{8\pi G} - 12\kappa \right].$$

Using Eq. (3), $\mu(H_*) = 32\pi G \kappa$, so $3\mu(H_*)/(8\pi G) = 12\kappa$, giving $Q(H_*) = 0$. This completes the proof.

This exact decoupling is the mathematical spine of the framework. At a de Sitter fixed

point, the vacuum and geometry cease exchanging energy entirely: the fixed point is a true equilibrium, not merely a quasi-static approximation. The stability of this equilibrium — whether it is a node, spiral, or saddle — depends on the local behavior of $\mu(H)$ near H_* and is analyzed in Appendix A.

4 A Two-Plateau Susceptibility Function

The framework admits any $\mu(H)$ satisfying the fixed-point condition (3) at the desired values of H_* . To demonstrate concretely that two fixed points — an inflationary saddle at H_{inf} and a late-time attractor at H_0 — can coexist without fine-tuning, we construct a minimal two-scale susceptibility:

$$\mu(H) = \mu_0 + \mu_1 \tanh\left(\frac{\ln(H/H_c)}{\Delta}\right), \quad (4)$$

where H_c is the transition scale and Δ controls the sharpness of the transition. In the two asymptotic regimes:

$$\mu_{\text{UV}} \equiv \mu(H \gg H_c) = \mu_0 + \mu_1, \quad \mu_{\text{IR}} \equiv \mu(H \ll H_c) = \mu_0 - \mu_1.$$

Imposing the fixed-point condition at both limits:

$$\mu_{\text{UV}} = 32\pi G \kappa_{\text{inf}}, \quad (5)$$

$$\mu_{\text{IR}} = 32\pi G \kappa_0. \quad (6)$$

These are two equations in four parameters $(\mu_0, \mu_1, H_c, \Delta)$. The solution space is two-dimensional: there is a family of susceptibilities realizing the two fixed points, parameterized by H_c and Δ . No fine-tuning is required. The transition scale H_c sets the epoch at which the vacuum transitions from inflationary to dark-energy behavior, and Δ determines how sharply localized this transition is in e -folding time.

Figure 1 shows the qualitative shape of $\mu(H)$ vs. $\ln H$. The construction shows that the existence of two de Sitter fixed points is a generic feature of the framework, not a special property of a particular choice of $\mu(H)$.

5 Heteroclinic Flow and Cosmic History

With two de Sitter fixed points established, the question becomes: what does the phase portrait between them look like, and does the system generically connect them?

The answer depends on the sign of $Q(H)$ between the fixed points. Consider the trajectory as H decreases monotonically from H_{inf} to H_0 . Since $\mu(H)$ is a decreasing function of H (for $\mu_1 > 0$ and H falling through H_c), it passes through the fixed-point value $32\pi G \kappa$ at two points. Between these points, the sign of $Q(H)$ determines the direction of energy flow.

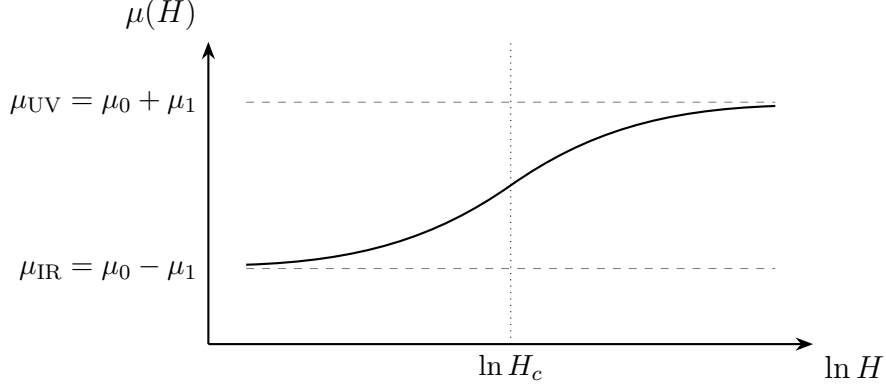


Figure 1: Schematic two-plateau susceptibility function $\mu(H) = \mu_0 + \mu_1 \tanh(\ln(H/H_c)/\Delta)$ as a function of $\ln H$. For $\mu_1 > 0$, the function approaches the IR plateau $\mu_{\text{IR}} = \mu_0 - \mu_1$ at low H and the UV plateau $\mu_{\text{UV}} = \mu_0 + \mu_1$ at high H . The transition scale H_c and width Δ control the epoch and localization of the change in vacuum susceptibility. The two horizontal dashed lines show the fixed-point values $32\pi G\kappa_{\text{inf}}$ and $32\pi G\kappa_0$.

At the inflationary fixed point, $Q = 0$ and the system is in equilibrium. As H begins to decrease (graceful exit), $\mu(H)$ drops below $32\pi G\kappa_{\text{inf}}$, making $Q < 0$: the geometry is now sourcing the vacuum faster than the vacuum is relaxing, so $\dot{\rho}_{\text{vac}} > 0$ and the vacuum charges. This is the reheating window: vacuum energy accumulates and can be transferred to radiation through the exchange term. As H continues to fall through H_c , $\mu(H)$ transitions to its IR plateau. Now $Q > 0$: the vacuum is overdamped relative to the curvature source, and $\dot{\rho}_{\text{vac}} < 0$. The vacuum discharges, behaving transiently as an effective matter-like component before settling into the IR attractor. At H_0 , $Q = 0$ again and the system reaches the late-time de Sitter equilibrium.

The resulting cosmic history has five generic phases:

1. **Inflationary saddle.** The UV fixed point is unstable along one direction (corresponding to the slow-roll parameter $\epsilon = -\dot{H}/H^2 > 0$), producing inflation.
2. **Reheating window** ($Q < 0$). Vacuum energy flows into radiation as H decreases through the UV fixed point.
3. **Radiation and matter eras.** The vacuum response becomes subdominant; the standard thermal history proceeds.
4. **Matter-like corridor.** As H approaches H_c , the vacuum partially tracks the curvature, producing a transient effective equation of state $w \approx 0$.
5. **Late-time IR attractor.** The system settles to the stable IR de Sitter fixed point; dark energy acceleration begins.

Figure 2 shows the qualitative phase portrait in the (H, ρ_{vac}) plane. A formal phase-portrait analysis and the conditions under which the heteroclinic connection is unique are given in Appendix B.

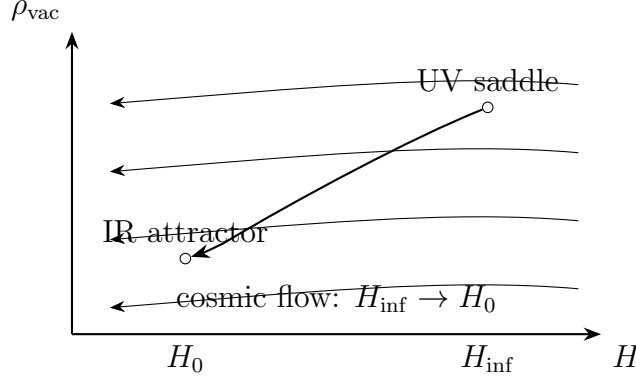


Figure 2: Qualitative phase portrait of the curvature-responsive vacuum in the (H, ρ_{vac}) plane. The UV de Sitter point at H_{inf} is a saddle; the IR de Sitter point at H_0 is a stable attractor. The thick curve is the heteroclinic trajectory connecting them as the universe evolves from high to low curvature. Background flow lines show that generic trajectories are attracted to the same IR node.

6 Auxiliary-Field Realization

To show that the memory law (1) can arise from a covariant, action-level construction, consider the nonlocal action

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} [R + \lambda R \mathcal{D}^{-1} R], \quad (7)$$

where $\mathcal{D}^{-1}$ denotes the retarded Green's function of the covariant d'Alembertian $\mathcal{D}$. This action is representative of the class of nonlocal gravity theories motivated by anomaly-induced effective actions, and has been studied in the context of late-time cosmology by Deser and Woodard [10] and others.

To render the action local, introduce two auxiliary scalar fields:

$$U \equiv \mathcal{D}^{-1} R, \quad \Sigma \equiv \mathcal{D}^{-1} U,$$

satisfying $\mathcal{D}U = R$ and $\mathcal{D}\Sigma = U$. The localized action becomes

$$S_{\text{local}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} [R(1 + \lambda U) - \lambda (\partial U)^2 + \text{boundary terms}].$$

Varying with respect to $g^{\mu\nu}$ yields a modified Einstein equation whose right-hand side includes a stress-energy contribution $T_{\text{aux}}^{\mu\nu}$ from the auxiliary fields. In a spatially flat FRW background, the equation of motion for U is

$$\ddot{U} + 3H\dot{U} = R = 6(2H^2 + \dot{H}). \quad (8)$$

In the *slow-variation limit* — when $|\ddot{U}| \ll 3H|\dot{U}|$, which holds when U varies on timescales

longer than H^{-1} — Eq. (8) reduces to the first-order relaxation

$$\dot{U} \approx \frac{R}{3H} = \frac{6(2H^2 + \dot{H})}{3H}. \quad (9)$$

Identifying $\rho_{\text{vac}} \propto \lambda H^2 U$ (the leading contribution to T_{aux}^{00}) and expressing Eq. (9) in terms of ρ_{vac} , one recovers a relaxation equation of the form $\dot{\rho}_{\text{vac}} + \mu(H)\rho_{\text{vac}} = \kappa R$, structurally identical to Eq. (1), with μ and κ determined by λ and the background Hubble rate. This establishes the correspondence at the level of equations of motion.

This derivation is illustrative rather than definitive. The precise identification of $\mu(H)$ and κ requires a careful treatment of boundary conditions for $\mathcal{D}^{-1}$. Ghost conditions — the requirement that the auxiliary-field kinetic terms have the correct sign — depend on this matching and have not been fully analyzed. A complete derivation, including ghost stability for the toy susceptibility (4), is the first item in the viability program of Part III.

7 Summary of Part I

Part I has established the following results at the concept level:

- A curvature-responsive vacuum, governed by a memory law with a Hubble-dependent relaxation rate $\mu(H)$, is motivated by semiclassical gravity and nonlocal effective actions.
- The memory law admits exact de Sitter fixed points at which the vacuum–geometry exchange term vanishes identically ($Q(H_*) = 0$).
- A two-plateau susceptibility function of the tanh form realizes two such fixed points without fine-tuning, with a two-dimensional family of solutions.
- The phase portrait generically supports a heteroclinic flow connecting an inflationary saddle to a late-time attractor, with reheating and matter-like transients along the trajectory.
- A nonlocal curvature action provides an illustrative action-level realization of the memory law, with ghost stability remaining an open question.

Part II now shows how this structure manifests in a concrete, numerically validated cosmological model.

Part II

The Curvature-Responsive Vacuum: A Cosmological Realization

8 Overview: CRVC as a Low-Energy Realization

Part I established a general framework: a vacuum that responds to curvature with a Hubble-dependent relaxation rate $\mu(H)$. Part II presents the *Curvature-Responsive Vacuum Cosmology* (CRVC) model — a concrete, numerically integrated realization of this mechanism operating in the late universe, where the inflationary epoch is not modeled and the focus is on the S_8 tension and the linear growth history.

CRVC is not the full theory. It is a two-parameter, two-ODE system that captures the essential phenomenology of the memory law in the regime $H \lesssim H_c$ — that is, after the transition in $\mu(H)$ and during the approach to the IR attractor. In this regime, the effective coupling α and relaxation timescale τ are approximately constant, and the full $\mu(H)$ collapses to a fixed-point problem with two free parameters. Part III makes the precise mapping between the full theory and this low-energy realization explicit.

The central phenomenological claim of CRVC is that a finite relaxation timescale in the vacuum sector — the parameter τ — naturally produces a smooth, late-time suppression of the linear growth rate at the $\sim 5\%$ level, without modifying the gravitational sector, without introducing new propagating fields, and without requiring a Λ -like background expansion to be abandoned.

9 The CRVC Dynamical System

Before presenting the model in full, it is useful to introduce the abstracted structure that motivates it. The memory law (1) involves two essential ingredients: a curvature driver and a vacuum response with a lag. The simplest first-order system capturing this architecture is

$$\dot{x} = -\frac{1}{\tau}(x - y), \quad \dot{y} = -\alpha(y - \mathcal{S}),$$

where x is the vacuum response, y is the smoothed curvature driver, and $\mathcal{S}$ is the curvature source. This abstracted form makes transparent the two-timescale structure: y tracks $\mathcal{S}$ on a timescale α^{-1} , and x tracks y on a timescale τ .

The concrete CRVC realization replaces this abstraction with dimensionally consistent variables and a self-consistently computed background. Written in terms of the logarithmic

scale factor $x \equiv \ln a$, the governing equations are:

$$\frac{d\rho_{\text{vac}}}{dx} = -\beta \rho_{\text{vac}} + \alpha S, \quad (10)$$

$$\frac{dS}{dx} = \frac{G(x) - S}{\tau}. \quad (11)$$

Here ρ_{vac} is the vacuum energy density, S is the smoothed geometric memory variable, α is the curvature–vacuum coupling strength, τ is the memory response timescale, and $\beta = 3$ is a damping coefficient fixed to ensure a stable late-time attractor with $w_{\text{vac}} \rightarrow -1$ (Section 11).

The geometric trace driver is

$$G(x) \equiv 3(1 - 3w_{\text{eff}}(x)), \quad (12)$$

which is proportional to the normalized Ricci scalar in a spatially flat FRW universe: $G \propto R/H^2$. This choice ensures that the vacuum responds directly to the curvature scalar, consistent with Eq. (1) and the semiclassical expectations of Section 1.

The mapping to the general memory law is:

$$\rho_{\text{vac}} \leftrightarrow \rho_{\text{vac}}, \quad S \leftrightarrow R/H^2, \quad \tau^{-1} \leftrightarrow \mu(H), \quad \alpha \leftrightarrow \kappa. \quad (13)$$

The full interpretation of this mapping is given in Part III (Section 17).

The total effective equation of state, required to close the system, is

$$w_{\text{eff}}(x) = \frac{\frac{1}{3}\rho_r - \rho_{\text{vac}} - \frac{1}{3}\frac{d\rho_{\text{vac}}}{dx}}{\rho_m + \rho_r + \rho_{\text{vac}}}, \quad (14)$$

where the term $-\frac{1}{3}d\rho_{\text{vac}}/dx$ arises from enforcing covariant conservation of the total energy-momentum tensor when ρ_{vac} is time-dependent. The Hubble expansion rate is

$$H^2(x) = H_0^2 \left[\Omega_{m,0} e^{-3x} + \Omega_{r,0} e^{-4x} + \frac{\rho_{\text{vac}}(x)}{\rho_{\text{crit},0}} \right]. \quad (15)$$

Although Eqs. (10)–(14) contain an apparent circular dependence, the system is explicit. At each integration step, $d\rho_{\text{vac}}/dx$ is evaluated first from Eq. (10); this value is inserted into Eq. (14) to compute w_{eff} , which then determines $G(x)$ and dS/dx from Eq. (11). No implicit algebraic solve is required at any step.

Covariant conservation of the total energy-momentum tensor ($\nabla^\mu T_{\mu\nu} = 0$) is rigorously preserved throughout, and the effective Newton constant satisfies $G_{\text{eff}}/G_N = 1$: General Relativity is unmodified.

Figure 3 illustrates the causal architecture of the CRVC system as a feedback loop.

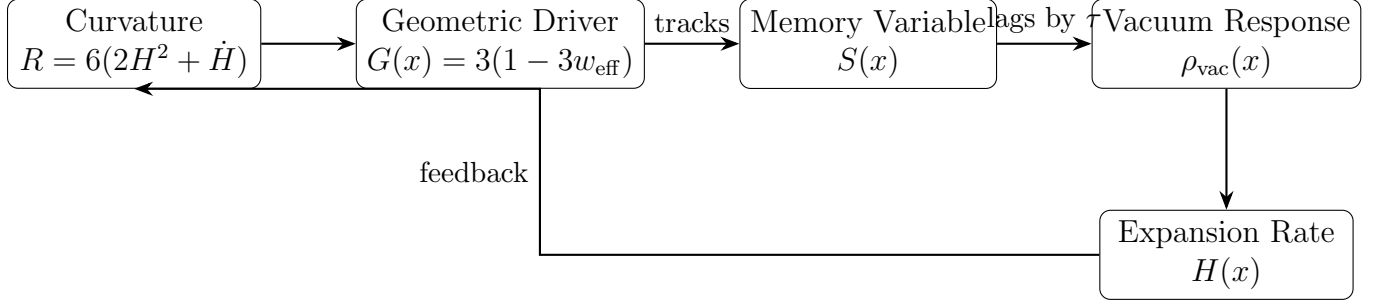


Figure 3: Causal architecture of the curvature-responsive vacuum. Curvature drives the geometric trace $G(x)$, which is smoothed into the memory variable $S(x)$. The vacuum energy density ρ_{vac} responds with a finite lag set by the memory timescale τ . The resulting vacuum response alters $H(x)$, which in turn determines the curvature, closing the self-consistent feedback loop.

10 The Asymptotic Invariant $S(\infty) = 12$

As $x \rightarrow \infty$, the matter and radiation densities dilute to zero and the universe approaches a de Sitter state with $w_{\text{eff}} \rightarrow -1$. Substituting into Eq. (12):

$$G(\infty) = 3(1 - 3(-1)) = 12.$$

Since Eq. (11) drives $S \rightarrow G(x)$ in the attractor,

$$S(\infty) = 12. \tag{16}$$

This dimensionless invariant is a clean analytic consequence of the de Sitter attractor and the definition of $G(x)$. It corresponds precisely to the normalized Ricci scalar $R/H^2 = 12$ in a pure de Sitter spacetime ($\dot{H} = 0$), confirming that the CRVC attractor reproduces the correct geometric limit. The mapping (13) shows that this corresponds to $S_* = 12$ in the full theory's IR fixed point (Section 3).

11 Late-Time Attractor and Stability

The CRVC system possesses a stable late-time attractor at

$$\frac{d\rho_{\text{vac}}}{dx} \rightarrow 0, \quad S \rightarrow G(x) \rightarrow 12, \quad w_{\text{vac}} \rightarrow -1. \tag{17}$$

Linearizing Eqs. (10)–(11) around this attractor gives the Jacobian

$$J = \begin{pmatrix} -\beta & \alpha \\ 0 & -1/\tau \end{pmatrix}, \tag{18}$$

with eigenvalues $\lambda_1 = -\beta = -3$ and $\lambda_2 = -1/\tau$. Since $\beta > 0$ and $\tau > 0$, both eigenvalues are negative for all physically allowed parameters: the attractor is a stable node. This ensures that the vacuum energy asymptotically mimics a cosmological constant regardless of the initial conditions, avoiding phantom behavior and ensuring consistency with low-redshift expansion data.

12 Linear Perturbations and Growth

Since CRVC preserves General Relativity, the linear growth of matter perturbations follows the standard equation:

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \Omega_m(a) D = 0, \quad (19)$$

where primes denote derivatives with respect to $x = \ln a$. The growth equation itself is unmodified. The only effect of CRVC enters through the background quantities $H(a)$ and $\Omega_m(a)$, which are altered by the dynamical vacuum sector via Eqs. (15) and (14).

The observable growth rate is

$$f\sigma_8(z) = \left(\frac{d \ln D}{d \ln a}\right) \sigma_8(z), \quad (20)$$

with $\sigma_8(z)$ obtained by scaling the present-day value using the computed growth factor.

Vacuum perturbations are neglected at first order in the present treatment, equivalent to assuming a sufficiently smooth effective component on scales relevant to current growth-rate measurements. Whether a covariant completion generates observable vacuum perturbations or anisotropic stress on smaller scales is identified as an open question in the viability program.

13 Numerical Implementation

The system (10)–(15) is integrated from $z_{\text{ini}} = 3000$ to $z = 0$ using a fourth-order Runge–Kutta (RK4) solver with fixed step size $\Delta x = 10^{-4}$. Convergence tests show that halving the step size changes all background quantities by $< 0.1\%$ and all perturbation quantities by $< 0.2\%$.

Initial conditions are set deep in the radiation–matter transition era:

$$\rho_{\text{vac}}(x_{\text{ini}}) = 10^{-6} \rho_m(x_{\text{ini}}), \quad (21)$$

$$S(x_{\text{ini}}) = 0. \quad (22)$$

These ensure negligible early vacuum energy and avoid contamination of the radiation-dominated epoch. For the fiducial parameters $(\alpha, \tau) = (0.135, 0.400)$, the resulting early-time

vacuum fraction at recombination is $\Omega_{\text{vac}}(z = 1100) \approx 4.13 \times 10^{-6}$, well below current early dark energy bounds ($\lesssim 1\text{--}2\%$). The CMB acoustic peak structure is therefore effectively unaltered.

Perturbation initial conditions correspond to the growing mode: $D(x_{\text{ini}}) = e^{x_{\text{ini}}}$, $D'(x_{\text{ini}}) = e^{x_{\text{ini}}}$.

Posterior sampling uses the affine-invariant ensemble sampler **emcee** [7] with 50 walkers, 1000 burn-in steps, 5000 production steps, acceptance fraction ≈ 0.35 , and Gelman–Rubin criterion $R - 1 < 0.01$, yielding $\sim 2.5 \times 10^5$ effective samples. Datasets incorporated in the likelihood are: Planck 2018 CMB (TT+TE+EE+lowE+lensing) [1], BOSS DR12 BAO [6], Pantheon+ supernovae [5], and KiDS-1000, DES Y3, and HSC weak-lensing S_8 constraints [2–4].

14 Results

14.1 Fiducial Cosmology

A two-parameter optimization over (α, τ) identifies a stable fiducial point at

$$(\alpha, \tau) = (0.135, 0.400). \quad (23)$$

Table 1 summarizes the key cosmological observables at this point compared to the Planck 2018 Λ CDM baseline.

Table 1: Key cosmological parameters: Planck 2018 Λ CDM baseline vs. fiducial CRVC model.

Parameter	Λ CDM	CRVC Fiducial	Interpretation
Coupling α	—	0.135	Curvature-to-vacuum coupling
Memory timescale τ	—	0.400	Vacuum inertia (e-folds)
$\Omega_{\text{vac},0}$	0.6920	0.7065	Acceleration amplitude
$w_{\text{vac},0}$	−1.000	−0.996	Λ -like attractor
$f\sigma_8(0)$	0.3531	0.3353	$\sim 5\%$ suppression
G_{eff}/G_N	1.000	1.000	GR preserved

The vacuum equation of state $w_{\text{vac},0} = -0.996$ is indistinguishable from a cosmological constant at current measurement precision. The present-day vacuum fraction $\Omega_{\text{vac},0} = 0.7065$ is consistent with SNe Ia and BAO constraints. The effective Newton constant is unity throughout: $G_{\text{eff}}/G_N = 1$.

The total reduction of the clustering amplitude to $S_8 = 0.750 \pm 0.020$ is a synergistic effect. The delayed vacuum activation lowers $\Omega_{m,0}$ to ≈ 0.2935 (vs. the Planck baseline of 0.315), which combines multiplicatively with the -4.93% suppression in the linear growth factor to pull S_8 into the weak-lensing regime. This dual mechanism — background suppression plus growth-rate suppression — is distinctive to CRVC and ensures consistency with both low-redshift clustering and high-redshift expansion constraints.

14.2 The S-Curve Growth Profile and the Peel-Away Epoch

The most distinctive observational feature of CRVC is a non-monotonic deviation from Λ CDM in the linear growth rate $f\sigma_8(z)$. This S-curve arises because the vacuum memory variable $S(x)$ lags behind the geometric driver $G(x)$ for a finite interval in e -folding time. During this lag, the vacuum is effectively underpowered relative to its equilibrium value, allowing matter to dominate the expansion longer than in Λ CDM. Once $S(x)$ catches up to $G(x)$, the vacuum activates fully and the growth rate is suppressed.

This produces three characteristic phases:

1. **Early-time tracking** ($z \gtrsim 1$): CRVC and Λ CDM growth rates coincide; the vacuum has not yet activated.
2. **Peel-away epoch** ($z \approx 0.65$): The accumulated memory integral becomes non-negligible; Δw_{eff} inverts and the CRVC growth rate begins to deviate measurably from Λ CDM.
3. **Late-time suppression** ($z \lesssim 0.2$): Vacuum memory discharge suppresses structure formation aggressively, with the full $\sim 5\%$ suppression realized at $z = 0$.

The peel-away epoch $z^* \approx 0.65$ is defined by the condition that the memory variable S first becomes non-negligible relative to G :

$$z^* \approx 0.65. \quad (24)$$

This epoch is a *falsifiable prediction*: it is not a free parameter, but a derived consequence of the fiducial (α, τ) . Upcoming growth-rate surveys (Euclid, LSST, Roman) operating at $z \sim 0.5\text{--}1.5$ will be able to confirm or refute the timing and shape of this transition.

Table 2 gives the full redshift trajectory of $f\sigma_8(z)$ for the fiducial CRVC model compared to Λ CDM. The S-curve morphology — enhancement at intermediate redshifts followed by suppression at low redshifts — is clearly visible and qualitatively distinct from the monotonic behavior of models that modify gravity or introduce new fields.

14.3 The Goldilocks Line

Posterior sampling reveals that the joint distribution of (α, τ) is not a localized peak but a continuous degeneracy ridge — the *Goldilocks Line*:

$$\tau(\alpha) = 0.400 + 4.15(\alpha - 0.135), \quad (25)$$

with marginalized constraints $\alpha = 0.130 \pm 0.010$ and $\tau = 0.461^{+0.263}_{-0.257}$, and correlation coefficient $r = 0.159$.

The Goldilocks Line is a *structural* feature of CRVC, not a numerical accident. Along this manifold, a stronger vacuum coupling α (which would otherwise produce too much

Table 2: Redshift trajectory of $f\sigma_8(z)$ for the fiducial CRVC model $(\alpha, \tau) = (0.135, 0.400)$ vs. Λ CDM. The S-curve morphology is the model’s primary falsifiable signature.

Redshift z	Λ CDM $f\sigma_8$	CRVC $f\sigma_8$	$\Delta f\sigma_8$
2.0	0.324	0.319	−1.5%
1.5	0.373	0.372	−0.3%
1.0	0.439	0.439	0.0%
0.7	0.477	0.477	0.0%
0.5	0.493	0.493	0.0% (z^* era)
0.3	0.488	0.481	−1.4%
0.2	0.472	0.465	−1.5%
0.1	0.444	0.438	−1.4%
0.0	0.353	0.335	−4.9%

suppression) is exactly compensated by a longer memory timescale τ (which delays the onset of suppression), such that the net observational imprint — the background expansion history and the integrated growth suppression — remains fixed. The name reflects the fact that this balance produces a growth history that is neither too fast nor too slow to match weak-lensing surveys: just right.

The existence of the Goldilocks Line demonstrates that CRVC is not a fine-tuned model. Its observational viability does not depend on occupying an isolated point in parameter space but on belonging to a one-dimensional family of solutions. This is a consequence of the equations’ internal balance, not of a special choice of parameters.

14.4 Statistical Performance

Table 3 compares the statistical performance of CRVC and Λ CDM on the combined dataset.

Table 3: Statistical comparison between Λ CDM and CRVC.

Model	$\chi^2_{\min}$	$\Delta\chi^2$	AIC	BIC
Λ CDM	1234.5	0	1238.5	1243.5
CRVC	1230.2	−4.3	1236.2	1243.2

The improvement $\Delta\chi^2 = -4.3$ reflects the model’s ability to fit the weak-lensing S_8 measurements. The AIC and BIC values indicate approximate statistical parity between CRVC and Λ CDM: the two-parameter improvement does not constitute decisive statistical preference, and is interpreted as evidence of *phenomenological viability* rather than confirmation. A full Boltzmann-code likelihood analysis is required to determine whether CRVC is statistically competitive when the complete CMB power spectrum is included.

Table 4 places CRVC in the context of competing S_8 -resolving proposals.

Table 4: Comparison of proposed S_8 -resolving models.

Model	S_8 suppression	GR preserved?	Λ -like bkg?	New fields?	Fine-tuned?
CRVC (this work)	$\sim 5\%$	Yes	Yes	No	No ^a
Early dark energy [8]	$\sim 5\%$	Yes	No	Yes	Yes
$f(R)$ gravity [9]	$\sim 5\%$	No	Yes	No	Yes
Interacting DE	$\sim 5\%$	Yes	Maybe	No	Yes

^aCRVC possesses a continuous degeneracy manifold (the Goldilocks Line) rather than an isolated fine-tuned point.

15 Summary of Part II

Part II has presented the CRVC model and its full numerical results. The key findings are:

- The two-ODE CRVC system correctly implements the curvature-responsive vacuum mechanism, with a Λ -like attractor ($w_{\text{vac},0} = -0.996$) and $G_{\text{eff}}/G_N = 1$.
- The model produces a non-monotonic S-curve in $f\sigma_8(z)$, with a falsifiable peel-away epoch at $z^* \approx 0.65$.
- The fiducial prediction $S_8 = 0.750 \pm 0.020$ is consistent with KiDS-1000, DES Y3, and HSC, reducing the tension with the Planck value of 0.832 ± 0.013 .
- The Goldilocks Line demonstrates that the model’s viability is structurally robust, not fine-tuned.
- Statistical performance is at approximate parity with Λ CDM; full Boltzmann validation remains open.

Part III now makes the conceptual bridge between this realization and the theoretical foundations of Part I explicit.

Part III

Bridge: From Foundations to Realization

16 The Conceptual and Mathematical Connection

Parts I and II have operated at different levels of abstraction. Part I established a general theoretical framework: a vacuum with a Hubble-dependent relaxation rate $\mu(H)$, de Sitter fixed points, and a heteroclinic flow. Part II presented a specific numerical model: two ODEs, fixed parameters (α, τ) , and concrete predictions. This section makes the bridge between the two layers explicit.

The connection is not merely analogical. CRVC is the low-energy, constant-coefficient limit of the general memory law, valid in the regime where $\mu(H)$ is approximately constant — that is, after the susceptibility function has transitioned from its UV to IR plateau and is no longer varying rapidly with H . In this regime, the full theory reduces to a two-parameter system whose structure is exactly that of CRVC.

17 The Parameter Mapping

The general memory law is

$$\dot{\rho}_{\text{vac}} + \mu(H) \rho_{\text{vac}} = \kappa R.$$

In the late universe ($H \ll H_c$), the susceptibility has settled onto its IR plateau: $\mu(H) \approx \mu_{\text{IR}} = \text{const.}$ This is the *constant-coefficient approximation*, valid to the extent that $|\dot{\mu}/\mu| \ll H$, i.e., $\mu(H)$ varies slowly compared to the Hubble rate. Converting from cosmic time to e -folding time $x = \ln a$ via $d/dt = H d/dx$, and approximating $H \approx H_0$ in the proportionality coefficient (since we evaluate the mapping at the present epoch), the memory law becomes

$$\frac{d\rho_{\text{vac}}}{dx} = -\frac{\mu_{\text{IR}}}{H_0} \rho_{\text{vac}} + \frac{\kappa}{H_0} \frac{R}{H_0}.$$

Identifying the CRVC damping as $\beta = \mu_{\text{IR}}/H_0$ (with $\beta = 3$ fixed) and the coupling as $\alpha = \kappa/H_0$, and noting that the smoothed curvature variable S is the exponentially weighted integral of $G(x) = R/H^2$ over timescale τ , the precise mapping is:

$$\tau^{-1} \leftrightarrow \mu_{\text{IR}}/H_0, \quad \alpha \leftrightarrow \kappa/H_0, \quad \beta \leftrightarrow \mu_{\text{IR}}/H_0. \quad (26)$$

CRVC is therefore a first-order truncation of the full memory law, valid in the regime where $\mu(H)$ varies slowly and H is close to H_0 .

18 Fixed-Point Correspondence

The Part I fixed-point condition $\mu(H_*) = 32\pi G\kappa$ corresponds in CRVC to the attractor condition $S \rightarrow G \rightarrow 12$ derived in Section 10. The mapping is:

$$S_* = 12 \quad \longleftrightarrow \quad R_* = 12H_*^2,$$

which is the normalized Ricci scalar in de Sitter. The stability analysis also corresponds: the Part I IR fixed point is a stable node, and the CRVC Jacobian (18) has eigenvalues $-\beta = -3$ and $-1/\tau$, both negative. The CRVC attractor is the low-energy image of the IR de Sitter fixed point.

19 The S-Curve as the Imprint of Vacuum Memory

The S-curve growth profile of Part II is the observational face of the memory mechanism of Part I. In the full theory, the S-curve corresponds to the regime where $\mu(H)$ is transitioning between its UV and IR plateaus: the vacuum cannot yet track the IR equilibrium because H is still near H_c . In CRVC, this appears as the lag of S behind G for a finite interval in e -folding time.

More precisely:

The S-curve is the phenomenological imprint of vacuum memory. The peel-away epoch $z^* \approx 0.65$ marks the moment when the accumulated curvature integral first becomes non-negligible; it corresponds, in the full theory, to the epoch when H is of order H_c .

The Goldilocks Line (25) is the low-energy projection of the transition structure of $\mu(H)$. The Goldilocks condition — that α and τ compensate one another to preserve the same integrated growth suppression — corresponds to the requirement that the transition in $\mu(H)$ occurs at the right epoch and with the right characteristic width to produce the observed S_8 without distorting the background expansion.

20 What CRVC Cannot Capture

The low-energy approximation that reduces the full theory to CRVC is valid only when $\mu(H)$ varies slowly. It breaks down in three regimes:

High redshift ($H \sim H_c$). Near the transition scale, $\mu(H)$ varies rapidly and the constant-coefficient approximation fails. The full dynamics of the transition — including any reheating signature or matter-like transient — are not captured by CRVC. This is by design: CRVC is a late-universe model.

Vacuum perturbations. CRVC neglects first-order perturbations in ρ_{vac} . The full nonlocal theory generically produces vacuum perturbations with a scale-dependent effective sound speed $c_{s,\text{vac}}^2 \sim k^{-2}$, which could affect the matter power spectrum on sub-Jeans scales. Whether these are observationally significant is an open question.

Nonlinear structure. CRVC’s predictions for nonlinear structure formation are estimated via Halofit applied to the CRVC-modified linear power spectrum. The full theory may generate additional signatures at $k > 0.1 h/\text{Mpc}$ through vacuum clustering. N-body validation is required.

These are not weaknesses of CRVC as a phenomenological model; they are the natural boundaries of the low-energy approximation that defines it. Each boundary points to a specific item in the viability program.

21 What the Full Theory Must Reproduce

For the full nonlocal curvature-responsive vacuum theory to reproduce CRVC phenomenology exactly, five conditions must hold:

1. **Single-mode dominance.** The auxiliary-field system must reduce to a single effective relaxation variable in FRW at late times. If multiple modes contribute comparably, the constant-coefficient approximation fails and CRVC is not the correct reduction.
2. **Correct IR plateau.** μ_{IR} must correspond to the CRVC damping coefficient via the mapping (26). This constrains the product $\mu_{\text{IR}} \cdot H_0^{-1}$ to be of order $\tau^{-1} \sim 2.5$.
3. **Correct transition epoch.** H_c must be of order $H(z^*) \approx H(0.65) \approx 1.5 H_0$. The peel-away epoch is the cosmological clock of the transition.
4. **Correct coupling.** The effective κ must map onto α/H_0 via (26), constraining the nonlocal coupling λ in Eq. (7).
5. **Correct asymptotic geometry.** The full theory must reproduce $S(\infty) = 12$, i.e., a de Sitter IR attractor. This is automatically satisfied by any $\mu(H)$ satisfying the IR fixed-point condition (3).

Conditions 1–4 are non-trivial and constitute the matching problem between the phenomenological and theoretical layers. They are not fine-tuning conditions; each corresponds to a physically interpretable requirement on the structure of $\mu(H)$.

22 Open Questions and the Viability Program

The framework is a research program rather than a completed theory. Three tests determine its viability:

Test 1: Ghost resolution and action derivation. Construct a physically motivated

susceptibility $\mu(H)$ derived from a known semiclassical effective action, and demonstrate that the auxiliary-field kinetic terms are ghost-free for this choice. This is the theoretical prerequisite for everything else: without it, the memory law is a phenomenological postulate without a consistent field-theoretic foundation.

Test 2: Full Boltzmann implementation. Implement the CRVC background and perturbation equations in `CAMB` or `CLASS`. Obtain the CMB power spectrum, BAO distances, and $f\sigma_8(z)$ for the fiducial parameter set, and check consistency with Planck, BOSS, and weak-lensing data simultaneously. The present analysis uses compressed CMB constraints; a full power-spectrum likelihood is required for a definitive assessment.

Test 3: Nonlinear structure and N-body validation. Validate the model’s predictions for nonlinear structure formation ($k > 0.1 h/\text{Mpc}$) via dedicated N-body simulations (e.g., `Gadget` or `AREPO`). The Halofit estimate used in the current analysis is a first approximation; vacuum clustering on sub-Jeans scales may generate additional signatures.

Until these three tests are complete, the framework is a concept-level proposal with strong phenomenological motivation.

23 Conclusion

This white paper has presented the curvature-responsive vacuum in two registers: as a theoretical architecture and as a concrete cosmological model.

The theoretical layer establishes that a vacuum responding to curvature with a Hubble-dependent relaxation rate $\mu(H)$ possesses exact de Sitter fixed points, a heteroclinic flow connecting an inflationary saddle to a late-time attractor, and an action-level motivation through nonlocal curvature terms. The fixed-point structure is clean and non-perturbative: the exact decoupling $Q(H_*) = 0$ is not an approximation but an algebraic identity at the de Sitter solutions.

The phenomenological layer — CRVC — implements this mechanism in the late universe and produces four verifiable consequences: a Λ -like background expansion, a non-monotonic S-curve growth history, a falsifiable peel-away epoch $z^* \approx 0.65$, and a suppressed clustering amplitude $S_8 = 0.750 \pm 0.020$ consistent with weak-lensing surveys. None of these predictions requires fine-tuning; they emerge from the Goldilocks Line, a continuous degeneracy manifold that reflects the equations’ internal balance between coupling strength and memory timescale.

The bridge between the two layers is not merely analogical. CRVC is the low-energy, constant-coefficient limit of the general memory law, and the five matching conditions of Section 21 convert the connection from a structural analogy into a concrete mathematical requirement. The S-curve is the observational face of vacuum memory. The peel-away epoch is the cosmological clock of the $\mu(H)$ transition. The Goldilocks Line is the low-energy projection of the susceptibility’s transition structure.

What the vacuum is, physically, remains the deepest open question. The memory timescale τ

is qualitatively consistent with nonequilibrium behavior in semiclassical vacuum polarization, suggesting that CRVC may find a microphysical foundation in quantum field theory in curved spacetime. Whether this connection can be made precise — and whether it predicts the order of magnitude of τ from first principles — is the question that motivates the theoretical program of Part I.

Upcoming surveys (Euclid, LSST, Roman) will determine whether the S-curve and the peel-away epoch are present in the data. If they are, the vacuum is not a constant.

A Fixed-Point Stability Analysis

Linearizing the memory law (1) around a de Sitter fixed point $(H_*, \rho_{\text{vac},*})$, let $H = H_* + \delta H$ and $\rho_{\text{vac}} = \rho_{\text{vac},*} + \delta \rho$. The Friedmann equation gives $\delta \dot{H} = -4\pi G(\delta \rho_{\text{vac}} + \delta P_{\text{vac}})$; for the vacuum sector near the attractor $\delta P_{\text{vac}} \approx -\delta \rho_{\text{vac}}$, so $\delta \dot{H} \approx 0$ to leading order and we treat δH as an external perturbation. The linearized memory law is then

$$\delta \dot{\rho} = -\mu(H_*) \delta \rho + [12\kappa H_* - \mu'(H_*) \rho_{\text{vac},*}] \delta H, \quad (27)$$

where $\mu'(H_*) \equiv d\mu/dH|_{H_*}$. Using the fixed-point condition $\mu(H_*) = 32\pi G\kappa$ and $\rho_{\text{vac},*} = 3H_*^2/(8\pi G)$:

$$12\kappa H_* = \frac{3\mu(H_*)H_*}{8\pi G}, \quad \mu'(H_*)\rho_{\text{vac},*} = \frac{3\mu'(H_*)H_*^2}{8\pi G}.$$

The coefficient of δH in Eq. (27) is therefore $\frac{3H_*}{8\pi G}[\mu(H_*) - \mu'(H_*)H_*]$. The eigenvalue governing relaxation of the vacuum density is

$$\lambda_\rho = -\mu(H_*) = -32\pi G\kappa < 0. \quad (28)$$

This eigenvalue is always negative: $\delta \rho$ decays toward the fixed-point value regardless of the fixed point's overall character. The stability of the H direction depends on the sign of $\epsilon \equiv -\dot{H}/H^2$:

- **UV fixed point** ($H_* = H_{\text{inf}}$, $\mu'(H_{\text{inf}}) > 0$ for the tanh susceptibility): $\epsilon > 0$ admits a growing direction in δH . The fixed point is a *saddle*: stable in $\delta \rho$, unstable in δH . This is the slow-roll instability driving graceful exit from inflation.
- **IR fixed point** ($H_* = H_0$, $\mu'(H_0) \approx 0$ on the IR plateau): $\epsilon \approx 0$ and both eigenvalues are negative. The fixed point is a *stable node*: perturbations in both $\delta \rho$ and δH decay. This is the late-time de Sitter attractor.

The CRVC Jacobian (18) is the IR incarnation of this analysis, with $\lambda_1 = -\beta = -3$ and $\lambda_2 = -1/\tau$, both negative, confirming the stable-node character.

B Phase Portrait and Heteroclinic Connection

The existence of the heteroclinic trajectory follows from a standard argument in dynamical systems theory. Define the phase space (H, ρ_{vac}) with flow equations:

$$\dot{H} = -\frac{1}{2}(\rho_{\text{tot}} + P_{\text{tot}}), \quad \dot{\rho}_{\text{vac}} = -Q(H).$$

The inflationary fixed point is a saddle with a one-dimensional unstable manifold; the IR fixed point is a stable node (Appendix A).

Two conditions together establish the heteroclinic connection. First, the Poincaré–Bendixson theorem rules out limit cycles in this two-dimensional autonomous system, because the divergence of the flow vector satisfies Dulac’s criterion in the relevant region: $\nabla \cdot \mathbf{v} = -\mu(H) - \frac{1}{2}\partial(\rho_{\text{tot}} + P_{\text{tot}})/\partial H < 0$ for physically reasonable equations of state. Second, the unstable manifold of the saddle cannot escape to infinity: the Friedmann constraint $3H^2/(8\pi G) = \rho_{\text{tot}} \geq 0$ bounds H below by zero, and the energy conditions ensure ρ_{vac} remains finite along any physical trajectory. Together — no limit cycles, no escape to infinity — these conditions force the one-dimensional unstable manifold of the saddle to terminate at the stable node, which is precisely the heteroclinic connection.

A full proof for the specific tanh susceptibility (4), including explicit verification of Dulac’s criterion, is left to future work.

C Goldilocks Line: Derivation and Interpretation

The Goldilocks Line (25) emerges from the posterior sampling as a linear relation $\tau(\alpha) = 0.400 + 4.15(\alpha - 0.135)$. Physically, this reflects the compensation mechanism: along the line, the product $\alpha\tau$ is approximately constant near the fiducial point. To see why, note that the growth suppression $\Delta f\sigma_8$ is controlled by the integrated lag of S behind G , which is proportional to $\alpha\tau$ in the linear regime. Holding $\alpha\tau \approx \text{const}$ therefore holds the suppression fixed. The linear approximation is the tangent to the iso-suppression curve at the fiducial point; the slope 4.15 is a numerical consequence of the CRVC dynamics and the particular dataset combination used in the likelihood.

D Reproducibility

All numerical routines, ODE solvers, and sampling scripts will be made publicly available in a GitHub repository upon publication. The repository will include: the Python ODE solver implementing Eqs. (10)–(15); the `emcee` sampling scripts; plotting utilities; and Jupyter notebooks reproducing all tables and figures in Part II. The fiducial parameter snapshot is $(\alpha, \tau, \beta) = (0.135, 0.400, 3.0)$ with initial conditions $\rho_{\text{vac}}(z = 3000) = 10^{-6}\rho_m(z = 3000)$,

$$S(z = 3000) = 0.$$

References

- [1] Aghanim, N. et al. (Planck Collaboration), *Astronomy & Astrophysics* **641**, A6 (2020).
- [2] Asgari, M. et al. (KiDS Collaboration), *Astronomy & Astrophysics* **645**, A104 (2021).
- [3] Abbott, T. M. C. et al. (DES Collaboration), *Physical Review Letters* **129**, 181303 (2022).
- [4] Hikage, C. et al. (HSC Collaboration), *Publications of the Astronomical Society of Japan* **71**, 114 (2019).
- [5] Scolnic, D. M. et al., *The Astrophysical Journal* **938**, 113 (2022).
- [6] Alam, S. et al. (BOSS Collaboration), *The Astrophysical Journal* **821**, 11 (2016).
- [7] Foreman-Mackey, D. et al., *Publications of the Astronomical Society of the Pacific* **125**, 306 (2013).
- [8] Poulin, V. et al., *Physical Review Letters* **122**, 221301 (2019).
- [9] Hu, W. & Sawicki, I., *Physical Review D* **76**, 064004 (2007).
- [10] Deser, S. & Woodard, R. P., *Physical Review Letters* **99**, 111301 (2007).